MTH 732 Topology - Homework 2

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Assignment 1. Show that if A is a strong deformation retract of X, then the inclusion $i : A \hookrightarrow X$ induces an isomorphism of fundamental groups.

Proof. We already know that i_* is a group homomorphism, so we just need to show that it is bijective. To show that it is surjective, let $F : X \times [0,1] \to X$ be a deformation retraction of X onto A, let $x_0 \in A$, and let $[\gamma]_X \in \pi_1(X, x_0)$. Now define $G : [0,1]^2 \to X$ by $G_t(a) \mapsto F_t(\gamma(a))$. Then $G_0 = \gamma$, $\operatorname{im} G_1 \subset A$, and $G_t(0) = G_t(1) = x_0$ since $x_0 \in A$ and $F_t|_A = \mathrm{id}_A$. Thus G is a path homotopy from γ to G_1, so $[\gamma]_X = [G_1]_X = i([G_1]_A)$, where $[G_1]_A \in \pi_1(A, x_0)$.

For injectivity, let $\alpha, \beta \in \pi_1(A, x_0)$ such that $i_*([\alpha]) = i_*([\beta])$. Then there exists a path homotopy $H : [0,1]^2 \to X$ such that $H_0 = \alpha$ and $H_1 = \beta$. We can then define $K = F_1 \circ H : [0,1]^2 \to A$, which is continuous since both H and the retraction F_1 are continuous. Moreover, $K_t(0) = H_t(0) = \alpha(0) = \beta(0)$ and $K_t(1) = H_t(1) = \alpha(1) = \beta(1)$ since F_1 is constant on A and the endpoint x_0 is in a. Finally, $K_0 = \alpha$ and $K_1 = \beta$, again since F_1 is constant on A. Thus, K is a path homotopy from α to β in A, so $[\alpha]_A = [\beta]_A$ as desired. □

Assignment 2. Let $p : E \to B$ be a covering map. Show that if E is path-connected and B is simply-connected[1], then p is a homeomorphism.

Proof. We know that p is surjective by definition, so we show that p is injective. Let $e_0, e_1 \in E$ such that $p(e_0) = p(e_1) = b_0$. Then there exists a path β connecting e_0 to e_1 since E is path-connected. Let $\alpha = p(\beta)$. Then $[\alpha] \in \pi_1(B, b_0)$, and since B is simply connected, we have a path homotopy $H : [0,1]^2 \to B$ from α to the constant path $f_{b_0} : [0,1] \to \{b_0\}$. This lifts to a path homotopy $\widetilde{H}$ from β to some path $\gamma : [0,1] \to p^{-1}(b_0)$ with $\gamma(0) = e_0$. Since $p^{-1}(b_0)$ is a discrete set, we must have that γ is actually constant, and thus is in particular the constant path at e_0. Since $\widetilde{H}$ is a *path* homotopy, however, it must fix endpoints, and thus $e_0 = e_1$. Therefore p is injective.

Now we need to show that p^{-1} is continuous. Let $x \in B$, let V be a neighborhood of $p^{-1}(x)$, and let U be an evenly covered neighborhood of x. The fact that p is a covering map tells us that $p|_{p^{-1}(U)}$ is a homeomorphism, so $(p|_{p^{-1}(U)})|_{p^{-1}(U)\cap V} = p|_{p^{-1}(U)\cap V}$ is also a homeomorphism, and thus its image $p(p^{-1}(U) \cap V) = U \cap p(V)$ is open since $p^{-1}(U) \cap V$ is open. Clearly $p^{-1}(U \cap p(V)) \subset V$, so we have continuity at x. Since x is arbitrary, we have continuity of p^{-1}, and thus it is a homeomorphism. □

[1] This means B is path-connected and has trivial fundamental group.

Assignment 3. Show that if $p_i : E_i \to B_i$, $i = 1, 2$, are covering maps then $p_1 \times p_2 : E_1 \times E_2 \to B_1 \times B_2$ is a covering map.

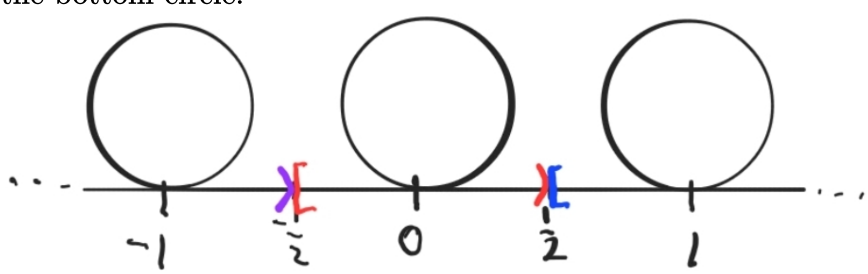
Proof. Let $(x, y) \in E_1 \times E_2$. Then there exist neighborhoods $N_1 \subset E_1$ and $N_2 \subset E_2$ of x and y respectively which are evenly covered under p_1 and p_2 respectively. Also, $p_1^{-1}(N_1) = \bigsqcup_\alpha U_\alpha$ and $p_2^{-1}(N_2) = \bigsqcup_\beta V_\beta$, where U_α and V_β are open for all α and β. Thus

$$(p_1 \times p_2)^{-1}(N_1 \times N_2) = \bigsqcup_{\alpha,\beta} U_\alpha \times V_\beta,$$

where each product $U_\alpha \times V_\beta$ is open by the openness of the factors, and the disjointness of the union comes from the fact that if $(a, b) \in (U_{\alpha_1} \times V_{\beta_1}) \cap (U_{\alpha_2} \times V_{\beta_2})$, then $a \in U_{\alpha_1} \cap U_{\alpha_2}$ and $b \in V_{\beta_1} \cap V_{\beta_2}$, so $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$. Also, $(p_1 \times p_2)|_{U_\alpha \times V_\beta}$ is a homeomorphism since it is the product of the homeomorphisms $p_1|_{U_\alpha}$ and $p_2|_{V_\beta}$. Thus $N_1 \times N_2$ is an evenly covered neighborhood of (x, y), and since (x, y) was arbitrary, $p_1 \times p_2$ is a covering map. $\square$

Assignment 4. Starting with the real line $\mathbb{R}$, glue a circle (like a wedge product) at every integer $m \in \mathbb{R}$. Call the resulting space E. Describe a covering map $p : E \to \mathbb{S}^1 \vee \mathbb{S}^1$.

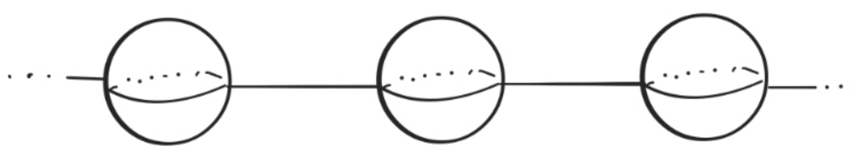
Proof. We will visualize $S^1 \vee S^1$ vertically, so that one circle is sitting atop the other. The circles in E will each map "identically" to the top circle in the wedge product. The half-open intervals (open on the right, that is) of length 1 centered at each wedge point in E will map to the bottom circle so that the two ends of the segment join at the bottom to "close" off the bottom circle.



$\square$

Assignment 5. Let $B = \mathbb{S}^2 \cup \{(x, 0, 0) : -1 \leq x \leq 1\}$. Find a simply-connected space E and describe a covering map $p : E \to B$.

Proof. We use the following covering space resembling a necklace of beads:



We map each S^2 to S^2 "identically" by "translation", i.e. we don't do anything special to the points on the spheres. Each line segment, modeled from left to right as $[-1, 1]$, is mapped in reverse order: $t \mapsto (-t, 0, 0)$. This way, the neighborhoods of points on a sphere not touching a line map locally homeomorphically to points on the sphere not touching the line in B, points "within" each line map locally homeomorphically to points "within" the line in

B, and the points where the lines and spheres touch map to the points where the line and sphere touch in B so that the neighborhoods are lines sticking out of discs (from the middle, on one side) as desired. $\square$